Math 54 — Week 5

Solutions

P1.

Idea. Interchange to get a leading 1, then use replacements only and count interchanges (Theorem 3).

1. One interchange, then clear column 1:

$$\begin{bmatrix} 2 & 4 & -2 & 6 \\ 1 & 3 & 0 & 2 \\ -1 & -1 & 4 & 1 \\ 3 & 8 & 1 & 5 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 3 & 0 & 2 \\ 2 & 4 & -2 & 6 \\ -1 & -1 & 4 & 1 \\ 3 & 8 & 1 & 5 \end{bmatrix} \xrightarrow[R_4 - 3R_1]{R_2 - 2R_1,\ R_3 + R_1} \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & -2 & -2 & 2 \\ 0 & 2 & 4 & 3 \\ 0 & -1 & 1 & -1 \end{bmatrix}$$

2. Clear column 2, then column 3:

$$\xrightarrow[R_4 - \frac{1}{2}R_2]{R_3 + R_2} \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & -2 & -2 & 2 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 2 & -2 \end{bmatrix} \xrightarrow{R_4 - R_3} \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & -2 & -2 & 2 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & -7 \end{bmatrix}$$

3. The diagonal product is $1 \cdot (-2) \cdot 2 \cdot (-7) = 28$, and there was one interchange, so $\det = -28$.

P2.

Idea. Use $\det(AB) = \det A \det B$, $\det A^{-1} = 1/\det A$, $\det A^T = \det A$ and $\det(cA) = c^n \det A$.

1. $\det(AB) = 3 \cdot (-2) = -6$.
2. $\det(A^{-1}) = \frac{1}{3}$ and $\det(A^T) = 3$.
3. $\det(2A) = 2^3 \cdot 3 = 24$, since A is 3×3.
4. By the product rule,

$$\det(B^T A^{-1} B) = \det(B^T) \det(A^{-1}) \det B = (-2) \cdot \frac{1}{3} \cdot (-2) = \frac{4}{3}.$$

P3.

Idea. Invertible exactly when $\det \neq 0$ (Theorem 4); compute det as a polynomial in h.

1. Expand along row 1:

$$\det = 1 \cdot \det \begin{bmatrix} 5 & 1 \\ 1 & 3 \end{bmatrix} - 2 \det \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} + h \det \begin{bmatrix} 2 & 5 \\ 0 & 1 \end{bmatrix} = 14 - 12 + 2h = 2(h+1).$$

2. So the matrix is invertible exactly when $h \neq -1$.

P4.

Idea. Cramer's rule: $x_i = \det A_i(\mathbf{b}) / \det A$.

1. $\det A = 5 \cdot 1 - 2 \cdot 3 = -1$, and
$$\det A_1(\mathbf{b}) = \det \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix} = -7, \qquad \det A_2(\mathbf{b}) = \det \begin{bmatrix} 5 & 1 \\ 3 & 4 \end{bmatrix} = 17.$$
2. So $x_1 = -7/(-1) = 7$ and $x_2 = 17/(-1) = -17$. Check: $35 - 34 = 1$ and $21 - 17 = 4$.

P5.

Idea. Move one vertex to the origin; the two adjacent sides are the columns. Then multiply by $|\det|$ of the map (Theorem 10).

1. Subtract $(-1, 0)$ from each vertex: they become $(0, 0)$, $(1, 5)$, $(2, -4)$, $(3, 1)$. Since $(1, 5) + (2, -4) = (3, 1)$, the sides at the origin are $(1, 5)$ and $(2, -4)$.
2. The area is
$$\left| \det \begin{bmatrix} 1 & 2 \\ 5 & -4 \end{bmatrix} \right| = |-4 - 10| = 14.$$
3. The map has determinant $2 \cdot 3 - 1 \cdot 1 = 5$, so the image has area $5 \cdot 14 = 70$.

P6.

Idea. Check $\mathbf{0}$ first; recognize null spaces and spans; otherwise test the three conditions.

1. (a) Yes: it is the null space of the 1×3 matrix $\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}$.
2. (b) No: $(1, 1)$ is in it, but $(1, 1) + (1, 1) = (2, 2)$ is not, since $2 \neq 2^2$.
3. (c) No: the zero polynomial has value 0 at 0, not 1.
4. (d) Yes: $(s - 2t, s, 3t) = s(1, 1, 0) + t(-2, 0, 3)$, so it is $\mathrm{Span}\{(1, 1, 0), (-2, 0, 3)\}$ (Theorem 1).
5. (e) Yes: $0^T = 0$; if $A^T = A$ and $B^T = B$ then $(A + B)^T = A^T + B^T = A + B$; and $(cA)^T = cA^T = cA$.

P7.

Idea. Row reduce $\begin{bmatrix} A & \mathbf{b} \end{bmatrix}$ once: the left part gives $\mathrm{Nul}\, A$, the last column decides $\mathbf{b} \in \mathrm{Col}\, A$.

1. Forward phase:
$$\left[\begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{array}\right] \xrightarrow[R_3 + 3R_1]{R_2 + 4R_1} \left[\begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & -2 & -6 & 5 \end{array}\right] \xrightarrow{R_3 - \frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{array}\right]$$
2. No row $\begin{bmatrix} 0 & 0 & 0 & b \end{bmatrix}$ with $b \neq 0$, so the system is consistent and $\mathbf{b}$ is in $\mathrm{Col}\, A$.
3. Backward phase on the coefficient part: $-\frac{1}{6}R_2$ gives $(0, 1, 3)$, and $R_1 + 3R_2$ gives $(1, 0, 5)$. So $A\mathbf{x} = \mathbf{0}$ means $x_1 = -5x_3$, $x_2 = -3x_3$, and $\mathrm{Nul}\, A = \mathrm{Span}\{(-5, -3, 1)\}$.
4. Check with the row-vector rule:
$$A \begin{bmatrix} -5 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 + 9 - 4 \\ 20 - 18 - 2 \\ 15 - 21 + 6 \end{bmatrix} = \mathbf{0}.$$

P8.

Idea. $\mathbf{u} \in \operatorname{Nul} A$: compute $A\mathbf{u}$. $\mathbf{v} \in \operatorname{Col} A$: row reduce $[\,A \;\; \mathbf{v}\,]$. Sizes decide the last part.

1. By the row-vector rule,
$$A\mathbf{u} = \begin{bmatrix} 6-8+2+0 \\ -6+10-7+0 \\ 9-14+8+0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \mathbf{0},$$
so $\mathbf{u}$ is not in $\operatorname{Nul} A$.

2. Row reduce $[\,A \;\; \mathbf{v}\,]$ (the second step scales row 3 by 2, which does not change consistency):
$$\left[\begin{array}{rrrr|r} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{array}\right] \xrightarrow[2R_3-3R_1]{R_2+R_1} \left[\begin{array}{rrrr|r} 2 & 4 & -2 & 1 & 3 \\ 0 & -1 & 5 & 4 & 2 \\ 0 & 2 & -10 & 9 & -3 \end{array}\right] \xrightarrow{R_3+2R_2} \left[\begin{array}{rrrr|r} 2 & 4 & -2 & 1 & 3 \\ 0 & -1 & 5 & 4 & 2 \\ 0 & 0 & 0 & 17 & 1 \end{array}\right]$$
The pivots are in columns 1, 2, 4 and not in the last column, so the system is consistent and $\mathbf{v}$ is in $\operatorname{Col} A$.

3. $\mathbf{u}$ has 4 entries and $\operatorname{Col} A$ is a subspace of $\mathbb{R}^3$, so $\mathbf{u}$ cannot be in $\operatorname{Col} A$.